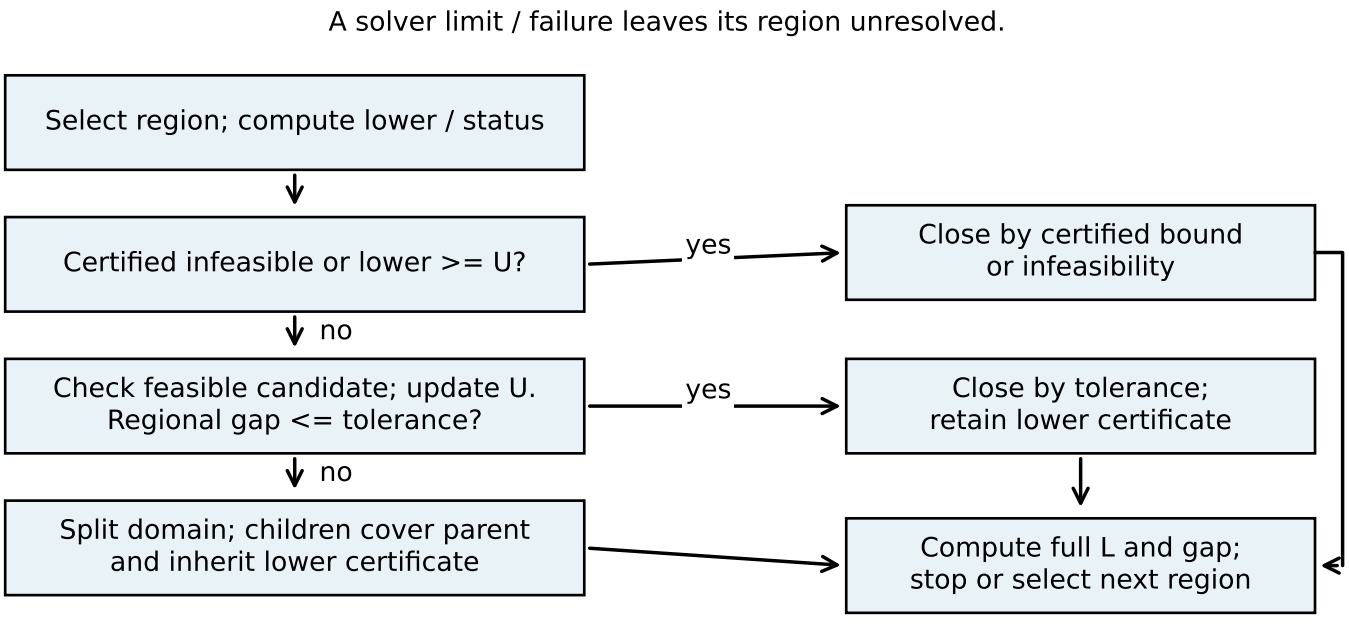
A solver limit / failure leaves its region unresolved.
Select region; compute lower / status
Certified infeasible or lower >= U?
yes
Close by certified bound or infeasibility
no
Check feasible candidate; update U. Regional gap <= tolerance?
yes
Close by tolerance; retain lower certificate
no
Split domain; children cover parent and inherit lower certificate
Compute full L and gap; stop or select next region